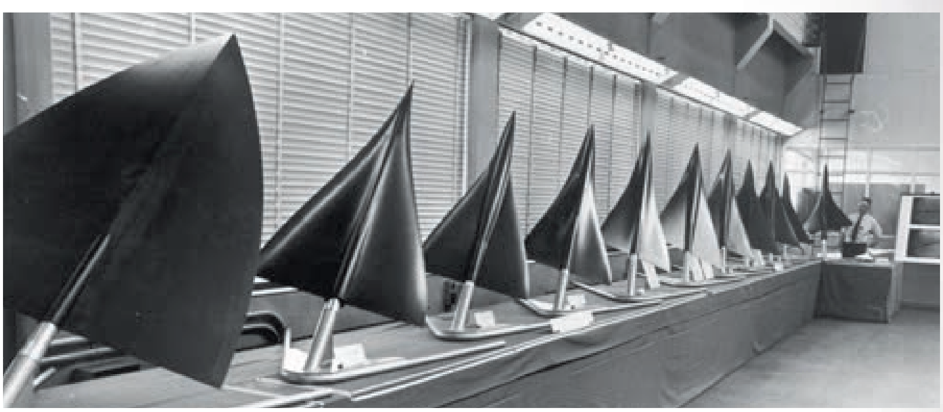
In the 1960s everyone in the airline business assumed that supersonic commercial jets would be the next big thing. The history of airliners seemed to show that increase in speed was a constant factor. In the 1930s the DC-3 had flown at 180 mph (290 kph); in the 1940s the four-engined propliners flew at upwards of 300 mph (480 kph); the turboprops flew at around 400 mph (640 kph) in the 1950s; and the Boeing 707 was carrying passengers at 600 mph (960 kph) in the 1960s. Why should this acceleration of commercial flight cease? It seemed a fair assumption that, given the choice, passengers would always opt for the shortest travel time, and thus the fastest airplane. As with turbojets, there were daunting problems to overcome, especially in the commercially crucial relationship between fuel consumption, payload, and range. But with the supersonic transport (SST) apparently the new holy grail for the aircraft industry, designers and engineers bent to the task.

Supersonic showdown

Three SST projects developed through the 1960s: the Anglo-French Concorde; the Soviet Tupolev Tu-144; and, in the US, the Boeing 2707. They were so costly that only government money could cover the expense—even Boeing depended almost entirely on federal funding. In many ways the SST projects were close in spirit to their contemporary, the American Apollo moon-landing program, driven by the same technological imperatives and heightened national pride, rather than by considerations of profit or practical advantage.

The Anglo-French and Soviet projects got under way more quickly, but the Americans were more ambitious, setting their sights on Mach 3 flight—around 2,000 mph (3,200 kph)—while their rivals settled for Mach 2. The Concorde and Tu-144 developed into such superficially similar, slender, delta-wing designs that in the West there were inevitable rumors of espionage. In fact, the number of possible solutions to the problem of designing an SST was very limited, and it is hardly surprising that two teams independently came up with a similar shape—no more surprising than that all three SSTs had a movable nose (to give better visibility on landing). Boeing pursued a radically different design, with a variable-geometry wing. It was not only intended to fly faster than its rivals, but also to carry more than twice as many passengers.

The Tu-144 made its first flight in December 1968, followed by Concorde in March 1969. Meanwhile, the Boeing SST was in trouble: the variable-wing concept had had to be abandoned and the projected passenger payload scaled down.



DELTA DESIGNS
Various shapes of delta wing suggested for Concorde are lined up on display. The one eventually chosen, furthest from the camera, had elegant curves that were aerodynamically complex, contrasting with the more simple angular delta adopted by the Tupolev bureau for the Soviet SST.